

Understanding CNNs

Convolutional Neural Networks for Astronomy

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Why astronomers use machine learning

Modern telescopes produce enormous image data sets.

Examples

- Classifying millions of galaxies.
- Finding faint gravitational lenses.
- Detecting supernovae and variable stars.
- Searching for exoplanet transit signals.



What is a CNN?

Simple definition

A Convolutional Neural Network, or CNN, is a neural network designed to learn patterns in images.

Instead of reading an image as one long list of numbers, a CNN looks at nearby pixels together.

A Convolutional Neural Network learns useful image features step by step



Images are numbers

A digital astronomy image is a grid.

$$\text{image} = \begin{bmatrix} 12 & 15 & 14 & 10 \\ 13 & 80 & 85 & 16 \\ 11 & 78 & 90 & 17 \\ 9 & 13 & 14 & 12 \end{bmatrix}$$

Each number stores brightness.

- Bright stars have large values.
- Empty sky has smaller values.
- Noise adds random variation.

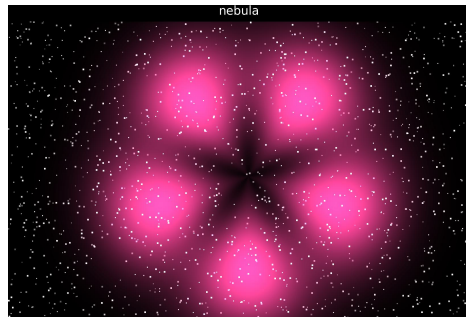
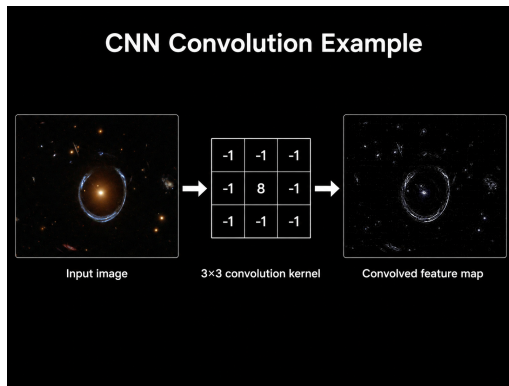


Image credit: artificial generated image

The key idea: convolution

A convolution is a small filter that slides across an image.



Intuition

Early CNN layers may learn simple patterns such as edges, blobs, gradients, and bright spots. Later layers combine these into objects such as spiral arms, galaxy cores, arcs, or star trails.

What does a CNN learn?

Early layers

- edges
- corners
- bright spots
- small textures

Later layers

- spiral arms
- galaxy bulges
- lensing arcs
- nebular filaments



Image credit: NASA/ESA Hubble

Example 1: classifying galaxies

A CNN can learn to classify galaxies as:

- spiral,
- elliptical,
- irregular,
- merger.

Why CNNs help

The model learns shapes directly from pixels instead of relying only on hand-made features.

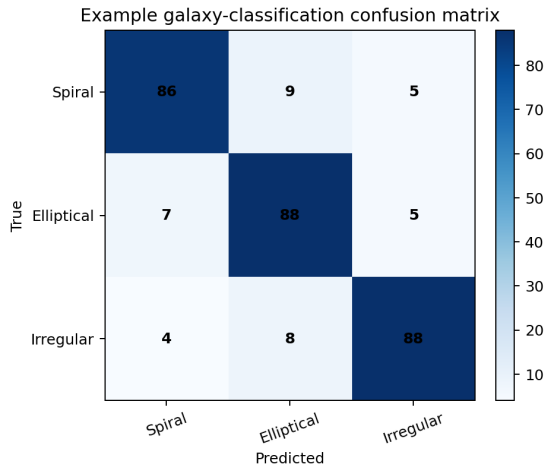


Image credit: NASA/ESA Hubble

Example galaxy-classification output

A confusion matrix shows where the model is correct and where it gets confused.

- Diagonal values are correct predictions.
- Off-diagonal values are mistakes.



Example 2: finding gravitational lenses

A gravitational lens can look like an arc or ring around a massive galaxy.

CNN task

Input: telescope image.

Output: probability that the image contains a lens.

This is useful because lenses are rare and large surveys contain millions of images.



Image credit: NASA. NASA copyright

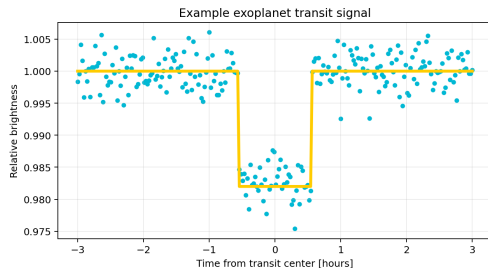
Example 3: exoplanet transits

When a planet crosses in front of a star, the star becomes slightly dimmer.

CNN or neural-network task

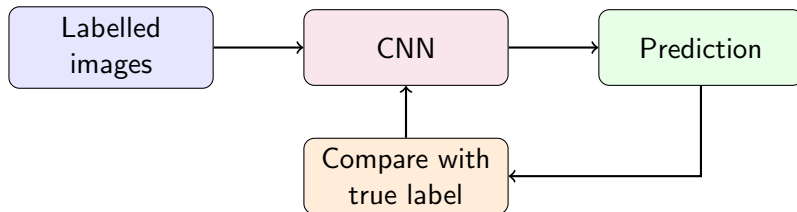
Input: light curve or image-like representation.

Output: planet candidate or non-candidate.



Training a CNN

A CNN learns from examples.



Learning

The model changes its filters again and again until its predictions improve.

A tiny CNN in Python

Keras-style example

```
Conv2D(32, (3,3), activation="relu"), MaxPooling2D((2,2)),  
Flatten(), Dense(64, activation="relu"), Dense(3, activation="softmax") ])
```

This could classify images into three classes: spiral, elliptical, and irregular.

Common beginner mistakes

- Training on too few images.
- Testing on images that are too similar to the training set.
- Ignoring class imbalance, for example many normal galaxies but few lenses.
- Forgetting that CNNs can learn telescope artifacts instead of astronomy.
- Reporting accuracy only, without checking false positives and false negatives.

Good practice in astronomy CNNs

- Split data into training, validation, and test sets.
- Use realistic noise and telescope conditions.
- Inspect misclassified examples.
- Compare with a simpler baseline method.
- Use explainability tools to see what image regions influenced the prediction.

Key message

A CNN is powerful, but astronomers must still check whether the model learned real astrophysics.

Take-home message

- ① CNNs are designed for image pattern recognition.
- ② Convolution filters learn local visual features.
- ③ Deeper layers combine simple features into complex astronomy structures.
- ④ CNNs are useful for galaxies, lenses, transients, and exoplanet searches.
- ⑤ Good validation is essential.



Thank you